

Teaching Exam

Subject – Computer Science

Topic – DBMS



Lecture No.- 2

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Topics to be Covered

Topic



- 1:- Database Schema
- 2:- Data Independence
- 3:- 3-Tier Schema Architecture
- 4:- Data Integrity
- 5:- Domain Integrity
- 6:- Entity Integrity
- 7:- Referential Integrity Constraints
- 8:- Conclusion



Topic : Database Schema

A database schema is a blueprint that defines the structure, organization, and relationships within a database. Basically It is Logical Representation of any Database

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Row 1				Rahul	12	A
Row 2				Pooja		
Row 3						

Key Components of a Database Schema:

- Tables: The fundamental structure for storing data.
- Columns: Represent individual pieces of data within a table.
- Relationships: Define how different tables are connected (e.g., primary and foreign keys).
- Constraints: Rules that ensure data integrity (e.g., unique constraints, data type restrictions).
- Data Types: Specify the type of data that can be stored in a column (e.g., text, numbers, dates).



Topic : Database Schema

Purpose of a Database Schema:

- **Organization:** Provides a structured way to organize and store data.
- **Data Integrity:** Ensures data accuracy and consistency through constraints and relationships.
- **Efficient Querying:** Facilitates efficient retrieval of data.
- **Data Modeling:** Serves as a blueprint for building the database.

Topic :Data Independence

Data independence in DBMS refers to the ability to make changes to one level of the database schema (logical or physical) without affecting the other levels or the applications that use the data.

Types of Data Independence:

1. Logical Data Independence:

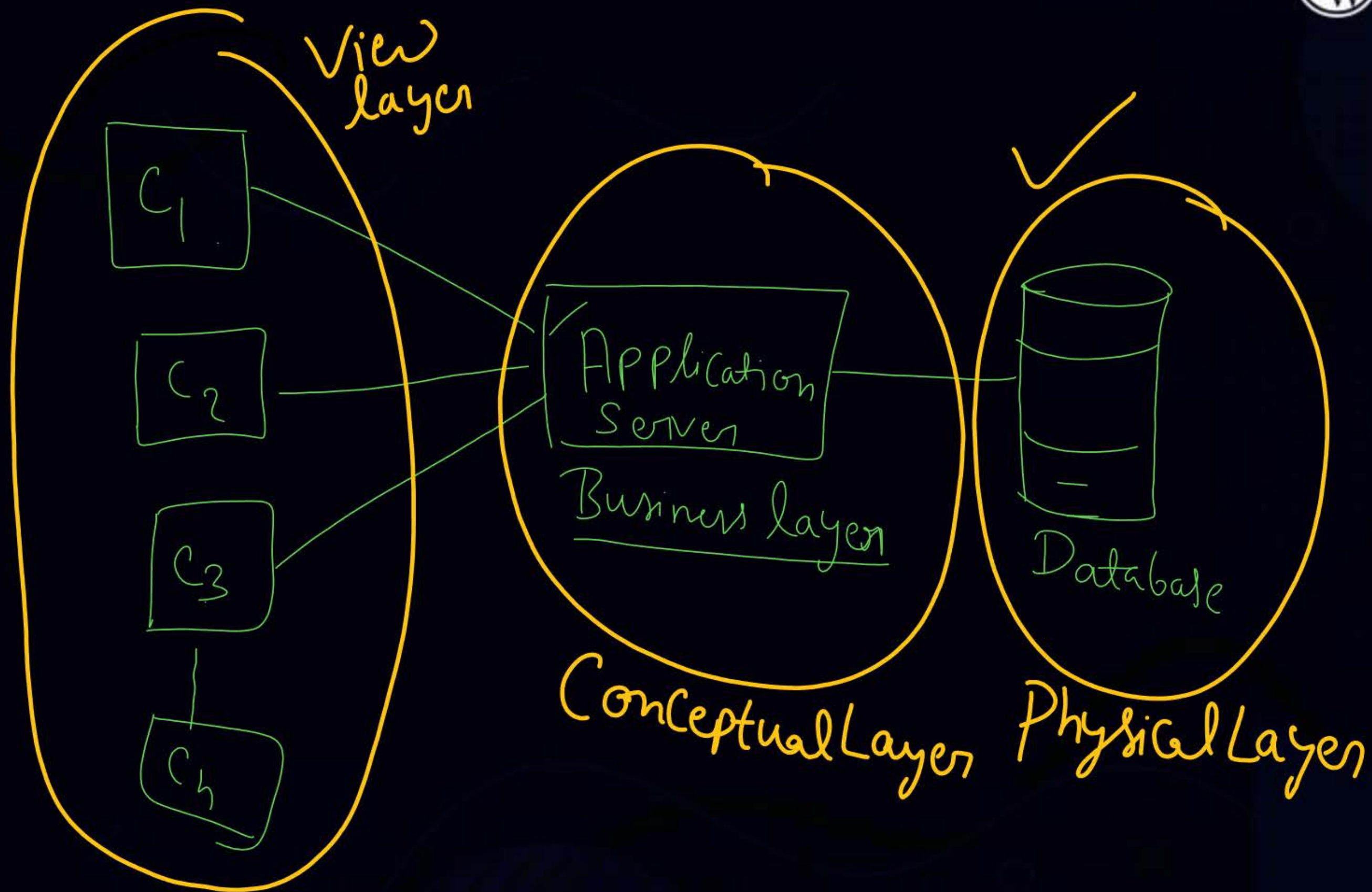
This allows changes to the conceptual schema (how the data is organized logically, e.g., adding or removing tables, changing data types) without requiring changes to the external schemas (how users view the data, e.g., views, reports) or application programs.

Example: Adding a new column to a table in the conceptual schema would not require changes to the user interface or application code that relies on that table.

2. Physical Data Independence:

This allows changes to the physical schema (how the data is physically stored, e.g., storage devices, indexing) without affecting the conceptual or external schemas or applications.

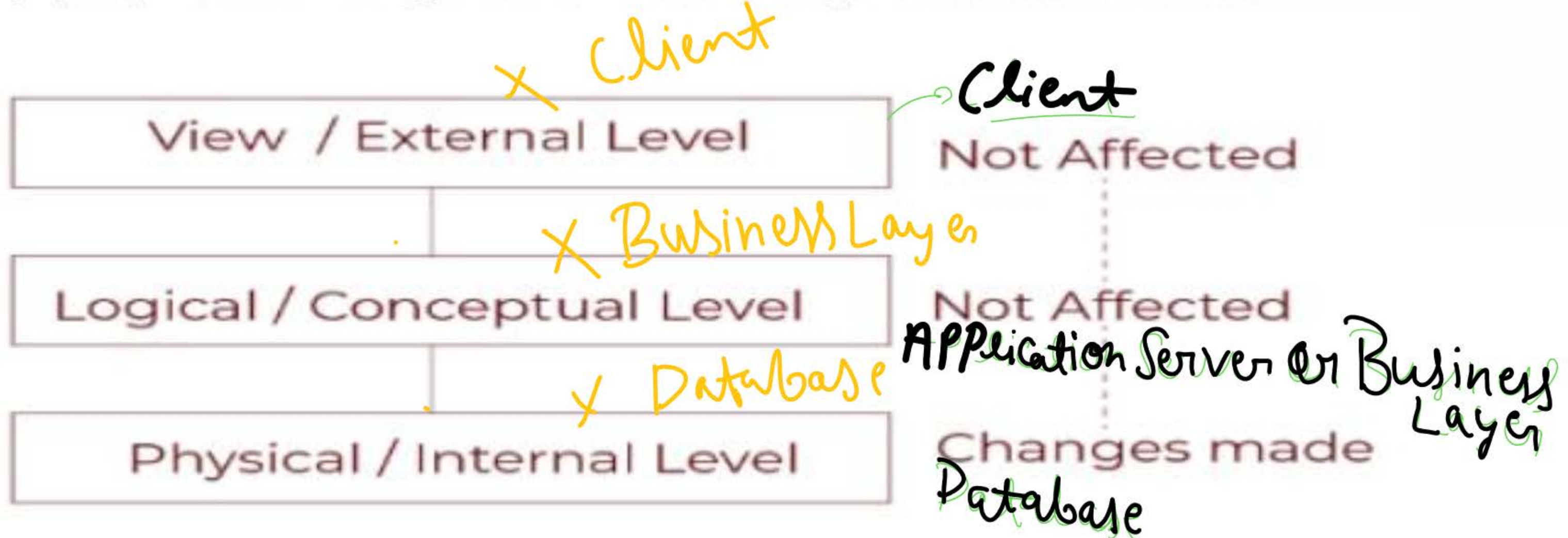
Example: Moving data to a different storage device or changing the indexing strategy would not require changes to application code or user interfaces.



Topic : Data Independence

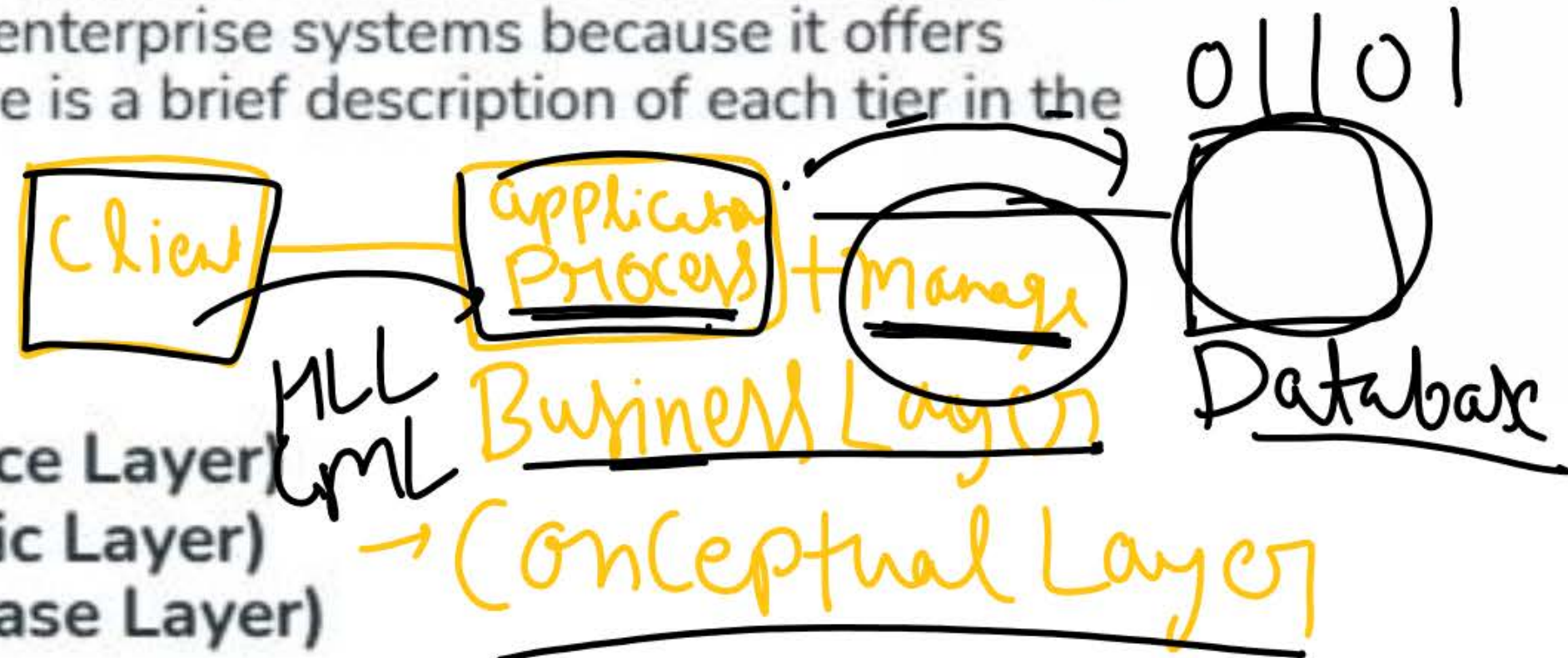


Data independence in DBMS refers to the ability to make changes to one level of the database schema (logical or physical) without affecting the other levels or the applications that use the data.



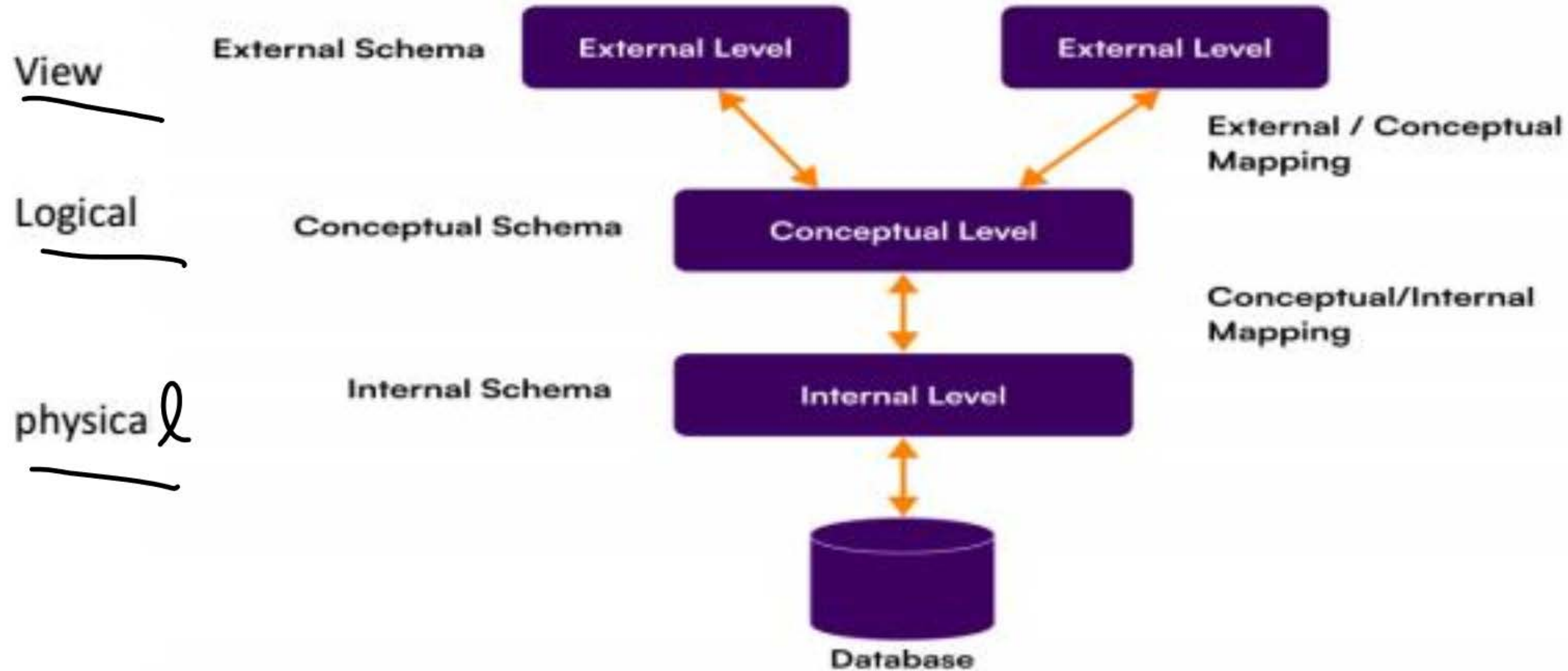
Topic :3-Tier Schema Architecture in DBMS

In DBMS, the 3-tier Schema architecture is a client-server architecture that separates the user interface, application processing, and data management into three distinct tiers or layers. The 3-tier architecture is widely used in modern web applications and enterprise systems because it offers scalability, flexibility, and security. Here is a brief description of each tier in the 3-tier architecture:



- 1-Presentation Tier (User Interface Layer)
- 2-Application Tier (Business Logic Layer)
- 3-Data Management Tier (Database Layer)

Topic :3-Tier Schema Architecture in DBMS



Topic :3-Tier Schema Architecture in DBMS

1:- External Level (View layer)

An external level specifies a view of the data in terms of conceptual level tables. Each external level view is used to cater to the needs of a particular category of users. For Example, FACULTY of a university is interested in looking course details of students, STUDENTS are interested in looking at all details related to academics, accounts, courses and hostel details as well. So, different views can be generated for different users. The main focus of external level is data abstraction.

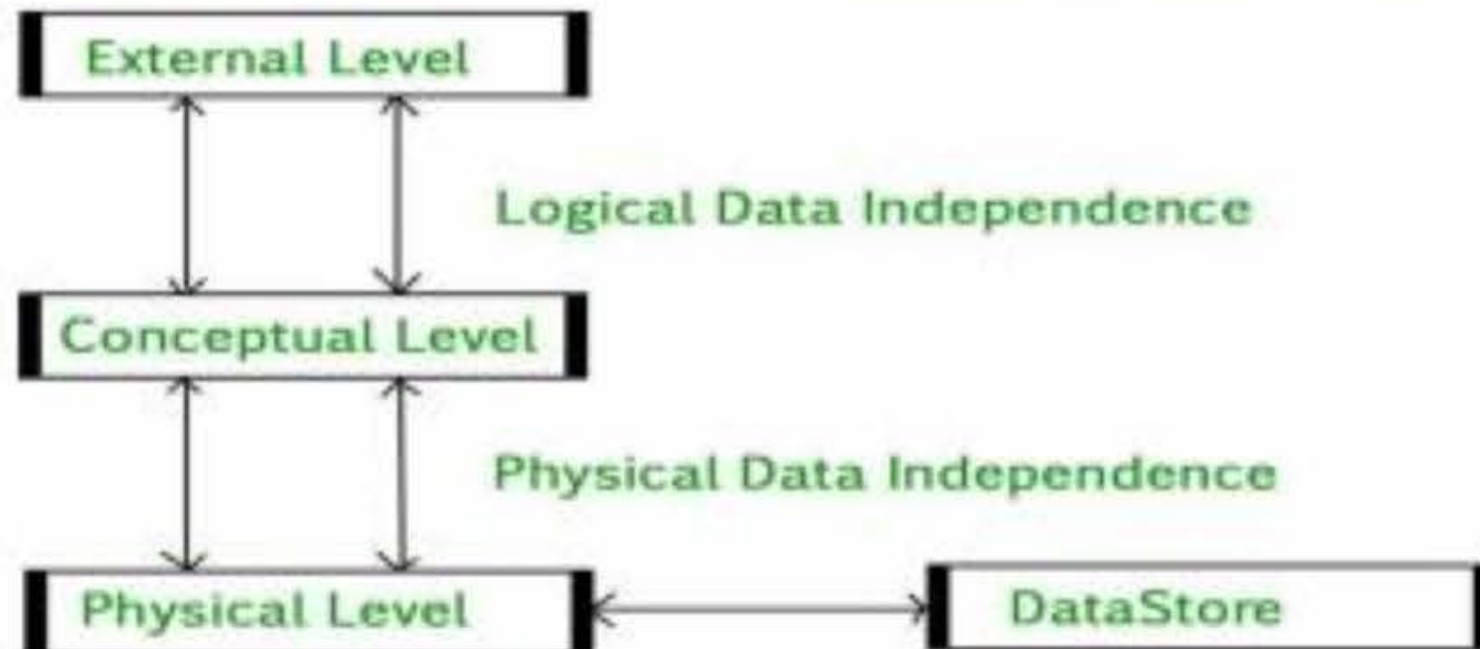
2:- Conceptual Level (Buffer Layer)

At conceptual level, data is represented in the form of various database tables. For Example, STUDENT database may contain STUDENT and COURSE tables which will be visible to users but users are unaware of their storage. Also referred as logical schema, it describes what kind of data is to be stored in the database.

Topic :3-Tier Schema Architecture in DBMS

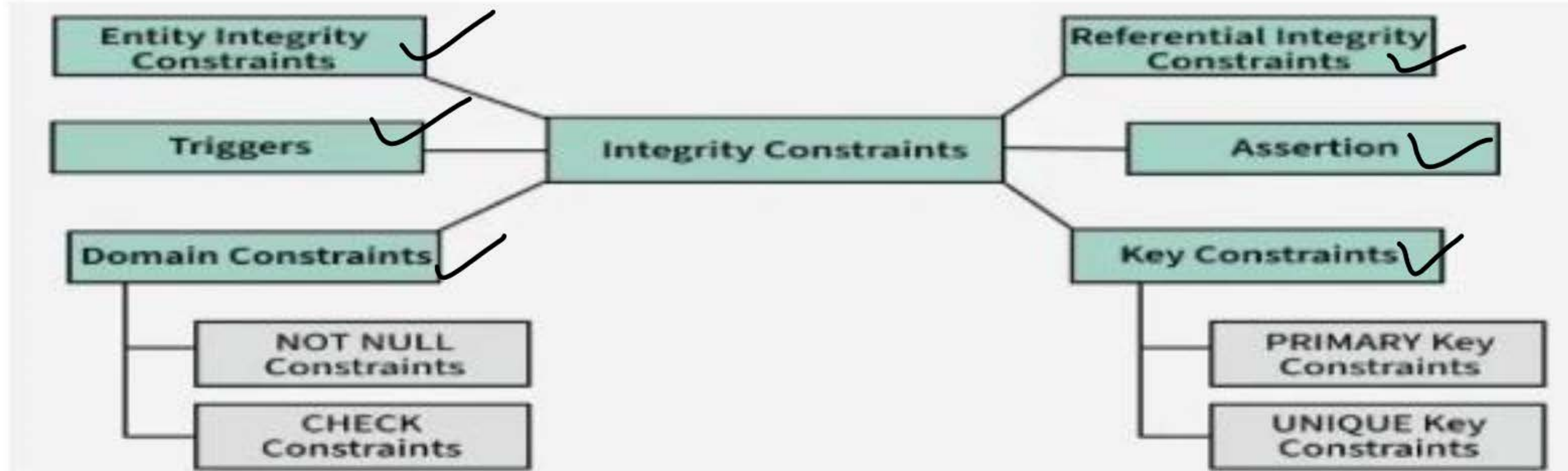
3. Physical Level

At the physical level, the information about the location of database objects in the data store is kept. Various users of DBMS are unaware of the locations of these objects. In simple terms, physical level of a database describes how the data is being stored in secondary storage devices like disks and tapes and also gives insights on additional storage details.



Topic :Data Integrity

Integrity constraints in a Database Management System (DBMS) are rules that help keep the data in a database accurate, consistent and reliable. They act like a set of guidelines that ensure all the information stored in the database follows specific standards.



Topic :Domain Integrity

Domain Integrity

Domain constraints are a type of integrity constraint that ensure the values stored in a column (or attribute) of a database are valid and within a specific range or domain. In simple terms, they define what type of data is allowed in a column and restrict invalid data entry. The data type of domain include string, char, time, integer, date, currency etc. The value of the attribute must be available in comparable domains.

Example:

Student_Id	Name	Semester	Age
21CSE100	Aniket Kumar	6th	20
21CSE101	Shashwat Dubey	7th	21
21CSE102	Manvendra Sharma	8th	22
21CSE103	Ashmit Dubey	5th	20

Varchar (Char)

Topic :Domain Integrity

This table demonstrates domain constraints in action by enforcing rules for each column:

1.Student_Id: Must be unique and follow a specific format like 21CSE###. No duplicates or invalid formats allowed.

2.Name: Accepts only valid text (no numbers) and cannot be left empty (NOT NULL constraint).

3.Semester: Allows specific values like 5th, 6th, etc., and ensures valid input (e.g., no 10th if not permitted).

4.Age: Must be an integer within a reasonable range (e.g., 18-30) and cannot contain invalid data like negative numbers or text

Topic :Entity Integrity

Entity Integrity Constraints

Entity integrity constraints state that primary key can never contain null value because primary key is used to determine individual rows in a relation uniquely, if primary key contains null value then we cannot identify those rows. A table can contain null value in it except primary key field.

Key Features of Entity Integrity Constraints

Uniqueness

- No duplicate entries are allowed in the primary key column.

NOT NULL

- The primary key column cannot contain NULL values, as every row must have a valid identifier.
- The primary key value must be unique for each row in the table.

Topic :Entity Integrity

Student_id	Name	Semester	Age
21CSE101	Ramesh	5th	20
21CSE102	Kamlesh	5th	21
21CSE103	Aakash	5th	22
NULL	Mukesh	5th	20

↗ Error

Primary Key ✓ ✓ ✓

Student_id	Name	Semester	Age
21CSE101	Ramesh	5th	20
21CSE102	Kamlesh	5th	21
21CSE103	Aakash	5th	22
21CSE102	Mukesh	5th	20

Topic :Entity Integrity



Why Key Constraints Are Important ?

- Prevent Duplicates: Ensure unique identification of rows.
- Maintain Relationships: Enable proper linking between tables (via foreign keys).
- Enforce Data Integrity: Prevent invalid or inconsistent data

Topic :Referential integrity constraints

foreign Key

Referential integrity constraints are rules that ensure relationships between tables remain consistent. They enforce that a Foreign key in one table must either match a value in the referenced primary key of another table or be NULL. This guarantees the logical connection between related tables in a relational database.

Why Referential Integrity Constraints Are Important ?

- Maintains Consistency: Ensures relationships between tables are valid.
- Prevents Orphan Records: Avoids cases where a record in a child table references a non-existent parent record.
- Enforces Logical Relationships: Strengthens the logical structure of a relational database

Topic : Referential integrity constraints

Here, in below example Block_No 22 entry is not allowed because it is not present in 2nd table.

Student_id	Name	Semester	Block_No
22CSE101	Ramesh	5th	20
21CSE105	Kamlesh	6th	21
22CSE102	Aakash	5th	20
23CSE106	Mukesh	2nd	22

Block_No	Block Location
20	Chandigarh
21	Punjab
25	Delhi

foreign Key

Primary Key

Topic :Assertion & Triggers

Assertion

An assertion is a declarative mechanism in a database that ensures a specific condition or rule is always satisfied across the entire database. It is a global integrity constraint, meaning it applies to multiple tables or the entire database rather than being limited to a single table or column.

Triggers

A trigger is a procedural statement in a database that is automatically executed in response to certain events such as INSERT, UPDATE, or DELETE. Triggers are often used to enforce complex integrity constraints or implement business rules that cannot be captured using standard constraints like primary keys or foreign keys.

Topic : Conclusion

Conclusion

Integrity constraints act as the backbone of reliable and robust database. They ensure that the data stored is reliable, worthy, consistent and accurate within the database. By implement integrity constraints we can improve the quality of the data stored in Database. So, as the database continues to grow it will not become inconsistent and inaccurate.

THANK YOU